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1. Prove that the set $S = \{\frac{p}{q} \mid p, q \in \mathbb{Z}, p > q > 0\}$ has no smallest member.

Proof. By contradiction, suppose $a \in S$ s.t

$$\begin{aligned} a &= \frac{p}{q} \\ \Rightarrow aq &= p \\ \Rightarrow q &= \frac{p}{a} \end{aligned}$$

is the smallest member in S . Let $p_0 = p + 1 \in \mathbb{Z}$ and $q_0 = q + 1 \in \mathbb{Z}$. Adding 1 to both sides,

$$\begin{aligned} p &> q > 0 \\ \Rightarrow p + 1 &> q + 1 > 1 > 0 \\ \Rightarrow p + 1 &> q + 1 > 0 \end{aligned}$$

satisfies the conditions for S . Furthermore, let $a_0 = \frac{p_0}{q_0} \in S$. We can rewrite a_0 to be

$$\begin{aligned} a_0 &= \frac{p + 1}{q + 1} \\ &= \frac{p + 1}{\frac{p}{a} + 1} \\ &= \frac{p + 1}{\frac{p}{a} + \frac{a}{a}} \\ &= \frac{p + 1}{\frac{p + a}{a}} \\ &= \frac{a(p + 1)}{p + a} \\ &= a \cdot \frac{p + 1}{p + a}. \end{aligned}$$

Since $p > q$,

$$\begin{aligned} \Rightarrow a &> 1 \\ \Rightarrow p + a &> p + 1 \\ \Rightarrow 1 &> \frac{p + 1}{p + a} \text{ where } p + a > 0. \end{aligned}$$

So, multiplying everything by a yields

$$a > \frac{a(p+1)}{p+a}$$

$$\Rightarrow a > a_0$$

which contradicts a being the smallest member in S , showing that we can always find a smaller member. $\therefore$ there must not be a smallest member in S . $\square$

2. (a) Suppose $a, b \in \mathbb{Z}$ with $a > b > 0$. Prove that the $\gcd(a, b) = \gcd(a - b, b)$.

Proof. Let $S_1 = \{n \mid a = nx, b = ny, n, x, y \in \mathbb{Z}\}$ be the set of all common divisors for a and b and $S_2 = \{m \mid a - b = ms, b = mt, m, s, t \in \mathbb{Z}\}$ be the set of all common divisors for $a - b$ and b . To show that the $\gcd(a, b) = \gcd(a - b, b)$, we need to show that $\forall$ elements in S_1 , they are also in S_2 and vice-versa. That is, we must show that $S_1 = S_2$. To that end, let $x \in S_1$. Since x is a divisor for both a and b , that means

$$\begin{aligned} a &= xy \quad \text{for some } y \in \mathbb{Z} \\ b &= xz \quad \text{for some } z \in \mathbb{Z}. \end{aligned}$$

Since x divides b , we only need to show that x also divides $a - b$. Taking the difference of a and b ,

$$\begin{aligned} a - b &= xy - xz \\ &= x(y - z). \end{aligned}$$

Given that $y, z \in \mathbb{Z}$, $k = y - z$ must also be an integer. So, by the definition of divisibility,

$$\begin{aligned} a - b &= xk \\ \Rightarrow x &\mid a - b \end{aligned}$$

x is also a divisor of $a - b$, meaning $x \in S_2$. Let $y \in S_2$. Since y is a divisor of both $a - b$ and b , that means

$$\begin{aligned} a - b &= ys \quad \text{for some } s \in \mathbb{Z} \\ b &= yt \quad \text{for some } t \in \mathbb{Z}. \end{aligned}$$

Since y divides b , we only need to show that y also divides a . Taking the sum of $a - b$ and b ,

$$\begin{aligned} (a - b) + b &= ys + yt \\ \Rightarrow a &= ys + yt \\ &= y(s + t). \end{aligned}$$

Since $s, t \in \mathbb{Z}$, $n = s + t$ must also be an integer. So, by the definition of divisibility,

$$\begin{aligned} a &= yn \\ \Rightarrow y &\mid a \end{aligned}$$

y is a divisor of a , meaning that $y \in S_1$. Since any element in S_1 is also in S_2 and vice-versa, it must be that $S_1 = S_2$. With their sets of common divisors being the same, we have shown that the $\gcd(a, b) = \gcd(a - b, b)$. $\square$

(b) Use the result from (a) repeatedly to simplify finding $\gcd(366, 240)$.

$$\begin{aligned} \gcd(366, 240) &= \gcd(240, 126) \\ &= \gcd(126, 114) \\ &= \gcd(114, 12) \\ &= \gcd(102, 12) \\ &= \gcd(90, 12) \\ &= \gcd(78, 12) \\ &= \gcd(66, 12) \\ &= \gcd(54, 12) \\ &= \gcd(42, 12) \\ &= \gcd(30, 12) \\ &= \gcd(18, 12) \\ &= \gcd(12, 6) \\ &= \gcd(6, 6) \\ &= \boxed{6} \end{aligned}$$

3. Suppose a and b are integers that each divide another integer c .

(a) If a and b are relatively prime, show that ab divides c .

Proof. Given that both a and b divide c , by the definition of divisibility,

$$\begin{aligned} c &= ax \quad \text{for some } x \in \mathbb{Z} \\ c &= by \quad \text{for some } y \in \mathbb{Z}. \end{aligned}$$

Consider a and b to be relatively prime, that is

$$\gcd(a, b) = 1.$$

By Bezout's Identity, $\exists$ some $s, t \in \mathbb{Z}$ s.t

$$as + bt = 1.$$

Multiplying both sides by c ,

$$\begin{aligned} cas + cbt &= c \\ \Rightarrow (by)as + (ax)bt &= c \\ \Rightarrow (ba)ys + (ab)xt &= c \\ \Rightarrow ab(ys + xt) &= c. \end{aligned}$$

Since y, s, x , and $t \in \mathbb{Z}$, $k = ys + xt$ must also be an integer. So,

$$\begin{aligned} c &= ab(k) \\ \Rightarrow ab &\mid c. \end{aligned}$$

□

(b) Give an example to show that if a and b are **not** relatively prime, then ab need not divide c .

Consider $a = 4$, $b = 8$, and $c = 24$

$\begin{aligned} \gcd(4, 8) &= 4 \neq 1 \text{ and both } 4 \mid 24 \\ &\text{and } 8 \mid 24, \text{ but } 32 \nmid 24 \end{aligned}$
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4. (a) Let $a, b, n \in \mathbb{Z}$ with $n > 1$. If $a \bmod n = a'$, and $b \bmod n = b'$, prove that

$$(a + b) \bmod n = (a' + b') \bmod n \text{ and } (ab) \bmod n = (a'b') \bmod n.$$

Proof. We must show that the remainder of $a + b$ is the same as the remainder of $a' + b'$ and that the remainder of ab is the same as the remainder of $a'b'$. By the Division Algorithm,

$$\begin{aligned} a \bmod n &= a' \\ \Rightarrow a &= cn + a' \text{ where } r = a', c \in \mathbb{Z} \end{aligned}$$

$$\begin{aligned} b \bmod n &= b' \\ \Rightarrow b &= dn + b' \text{ where } r = b', d \in \mathbb{Z} \end{aligned}$$

Taking the sum of a and b ,

$$\begin{aligned} a + b &= cn + a' + dn + b' \\ &= cn + dn + a' + b' \\ &= (c + d)n + (a' + b') \text{ where } r = a' + b' \end{aligned}$$

Although both a' and b' are less than n , their sum could exceed it and fail the Division Algorithm. By the Division Algorithm, let $q_0, r_0 \in \mathbb{Z}$ s.t

$$a' + b' = q_0n + r_0.$$

Plugging this expression back into $a + b$,

$$\begin{aligned} a + b &= (c + d)n + q_0n + r_0 \\ &= (c + d + q_0)n + r_0 \quad \text{where } r = r_0 \end{aligned}$$

Since $r = r_0$ for both $a + b$ and $a' + b'$, $(a + b) \bmod n = (a' + b') \bmod n$. Finding the product of a and b ,

$$\begin{aligned} ab &= (cn + a')(dn + b') \\ &= cndn + cnb' + a'dn + a'b' \\ &= (cdn + cb' + a'd)n + a'b' \quad \text{where } r = a'b' \end{aligned}$$

But, the product of $a'b'$ could also exceed n and violate the Division Algorithm. By the Division Algorithm, let $p_0, t_0 \in \mathbb{Z}$ s.t

$$a'b' = p_0n + t_0.$$

Plugging this expression back into ab ,

$$\begin{aligned} ab &= (cdn + cb' + a'd)n + p_0n + t_0 \\ &= (cdn + cb' + a'd + p_0)n + t_0 \quad \text{where } r = t_0 \end{aligned}$$

Since $r = t_0$ for both ab and $a'b'$, $(ab) \bmod n = (a'b') \bmod n$. □

(b) Use the result from part (a) to simplify the calculations of $(51 + 68) \bmod 7$ and $(82 \cdot 73) \bmod 7$ as much as possible.

$$51 \bmod 7 = 2 \qquad 68 \bmod 7 = 5$$

$$\begin{aligned} \Rightarrow (51 + 68) \bmod 7 &= (2 + 5) \bmod 7 \\ &= \boxed{0} \end{aligned}$$

$$82 \bmod 7 = 5 \qquad 73 \bmod 7 = 3$$

$$\begin{aligned} \Rightarrow (82 \cdot 73) \bmod 7 &= (5 \cdot 3) \bmod 7 \\ &= \boxed{1} \end{aligned}$$